

Redocking Validation Reveals Structure-Dependent Docking Reliability in ATR: A Reproducible Pilot Virtual Screen of BRICS-Generated Candidates

[Author name(s) to be added]

[Affiliation to be added]

Correspondence: [email to be added]

Abstract

Background: Nearly all existing chemical probes of the ATR–CHK1 replication-stress checkpoint are ATP-competitive kinase inhibitors. Two 2025 cryo-EM structures of the ATR kinase domain (PDB 9L40, 2.87 Å, bound to berzosertib/VE-822; PDB 9L4B, 3.20 Å, bound to camonsertib/RP-3500) provide new templates for structure-based work on this target, and companion depositions revealed berzosertib additionally engaging an allosteric site at the ATR–ATR dimer interface, distinct from the canonical ATP pocket. Whether a newly deposited structure is reliable for conventional ATP-pocket docking is not something resolution alone can guarantee. **Methods:** We assembled a small seed library of five ATR/CHK1-directed compounds, three extracted directly from the co-crystallized active-site ligand of PDB 9L40, 9L4B, and a CHK1 reference structure (PDB 2YM8), generated derivatives by BRICS recombination, filtered for developability by standard descriptor rules, and – before docking any candidate – validated each receptor by redocking its own native ligand (naïve and symmetry-corrected RMSD, three seeded AutoDock Vina replicates) against a pre-specified 2.0 Å pass threshold. **Results:** Redocking passed for 9L4B (0.61 ± 0.02 Å) and 2YM8 (0.76 – 0.90 Å) but reproducibly *failed* for 9L40 (naïve 2.96 ± 0.06 Å, symmetry-corrected 2.53 ± 0.01 Å – still over threshold) despite looser-than-passing but still self-consistent search reproducibility (0.56 Å); notably, 9L40 is the *higher*-resolution structure of the two. Two alternative explanations were tested and ruled out: the docking box cannot reach the nearby allosteric site (60.65 Å away), and the deposited pose is not an outlier by map-model Q-score (though residue inclusion is somewhat lower). 9L40 was excluded from scoring per the pre-specified rule. A 40-compound sanity-check screen (5 seeds + 35 developability-passed derivatives) against the validated 9L4B structure recovered camonsertib – the receptor’s own ligand – as top-ranked, with no derivative exceeding the seed compounds. **Conclusions:** A newly deposited, high-resolution structure is not automatically a reliable docking receptor; this can only be established empirically, per structure and per protocol, before any candidate is scored. This n=1 finding motivates a larger survey using the same pipeline. All code, data, and structures are openly available.

Keywords: ATR kinase; molecular docking; redocking validation; BRICS; virtual screening; reproducibility; AutoDock Vina

Background

The ATR–CHK1 checkpoint kinase axis is the principal sensor of DNA replication stress: ATR is activated at stalled replication forks through protein-mediated mechanisms (the ATRIP–TOPBP1 axis, and independently ETAA1) rather than by occupation of its own ATP-binding pocket, and once active it phosphorylates and activates CHK1, which restrains further origin firing and protects replication forks from nucleolytic collapse (1–3). Genetic and pharmacological attenuation of ATR or CHK1 activity increases instability at common fragile sites – chromosomal loci prone to breakage under replication stress – consistent with this axis being protective (4, 5).

Essentially all existing chemical matter directed at this pathway – berzosertib, ceralasertib, camonsertib, prexasertib, and related compounds – is ATP-competitive kinase inhibitors developed for oncology synthetic-lethality strategies (6). Two cryo-EM structures of a homodimer of the ATR kinase domain released in 2025 – bound respectively to berzosertib (PDB 9L40, 2.87 Å) and camonsertib (PDB 9L4B, 3.20 Å) – are a substantial improvement over the original 2018 structure of the full ATR–ATRIP complex (PDB 5YZ0, 4.7 Å), and, together with companion depositions of the complete ATR–ATRIP assembly from the same study, newly resolved that one such assembly engages *four* berzosertib molecules – two in the canonical ATP-competitive active sites, and two at a previously uncharacterized allosteric pocket formed at the ATR–ATR dimer interface (7). This is a structurally more complex picture than a single ATP-pocket story, and it raises a practical question for anyone using these new structures for docking: does a newly deposited, high-resolution structure automatically make a trustworthy receptor for conventional ATP-pocket virtual screening? As will become apparent below, the answer this pilot finds is no – and, notably, it is the *higher*-resolution of the two structures (9L40, 2.87 Å) that fails.

This is an empirical question, not one that follows from resolution or recency, and it is the question this pilot study is organized around. Before screening any candidate against 9L40 or 9L4B, we validate each structure’s suitability as a docking receptor by redocking its own co-crystallized ligand – the standard control for docking-protocol accuracy, following the convention established on the Astex Diverse Set (8) and implemented here with AutoDock Vina (9) – and only proceed to score novel candidates against a structure that passes. We then report a small pipeline sanity-check screen of BRICS-derived (10) candidates against whichever structure(s) pass, as a demonstration of the validate-before-screening principle applied to a concrete, currently topical target.

Scope

This is explicitly a pilot study, not a comprehensive virtual screening campaign. Docking, by itself, estimates predicted binding and cannot establish whether a ligand activates, inhibits, or otherwise modulates ATR or CHK1 – physiological ATR activation depends on upstream protein interactions absent from a static structural model (1, 2). All results below are reported as predicted binding and prioritization, not as demonstrated pharmacology. Molecular dynamics and MM-GBSA relative energetic estimation, and a broader cross-kinase selectivity panel, were part of the original design of the computational pipeline this pilot draws on, but are not included here: they require substantially more compute (GPU-accelerated MD) and additional validated receptor structures than were available for this pilot, and are explicitly left as future work (see Limitations) rather than run at a scale too small to be informative.

Methods

Receptor structures

Three receptor structures were obtained, in mmCIF format, via the Registry of Open Data on AWS mirror of the RCSB PDB archive ([s3://pdbsnapshots](https://pdbsnapshots.s3.amazonaws.com/), snapshot 2026-01-01), since the primary RCSB PDB web/FTP endpoints were not reachable from the compute environment used for this study: PDB 9L40 (ATR kinase domain, homodimer, bound to VE-822/berzosertib, 2.87 Å), PDB 9L4B (ATR kinase domain, homodimer, bound to camonsertib/RP-3500, 3.20 Å), and PDB 2YM8 (human CHK1 kinase domain bound to a chloroisoquinolinylamino-pyrazine-carbonitrile inhibitor, 2.07 Å) (7). The original 2018 ATR–ATRIP structure (PDB 5YZ0, 4.7 Å overall) was not used as a docking receptor: at that resolution, ATP-pocket side-chain positions are not reliably resolved.

For each structure, *both* protein chains (the full homodimer; standard residues only, via Biopython (11)) were retained as the docking receptor, and the co-crystallized ligand of the biologically relevant

ATP-competitive active site was identified and isolated. For 9L40, four copies of the ligand (chemical component A1EIE) are present per asymmetric unit; contact analysis (heavy atoms within 4.5 Å of each protein chain) confirmed that residues A2701/B2701 each contact only their own chain (canonical, single-monomer ATP-competitive site) while A2702/B2702 contact both chains (the dimer-interface allosteric site reported by Wang et al. (7)); A2701 was used as the active-site reference ligand. The centroid-to-centroid distance from A2701 to the nearest allosteric copy (A2702) is 60.65 Å – far outside the reach of the 22 Å docking box used here (Table 2 below), which rules out the empty allosteric cavity as a confound for the redocking result that follows: the docking box could not have reached it. The analogous single-copy active-site ligands (chemical component A1EIK for 9L4B; YM8 for 2YM8) were extracted directly. New-style five-character PDB chemical-component codes overflow the fixed-width residue-name column of the legacy PDB format and silently corrupt naive PDB-to-PDBQT conversion (each atom is parsed as an isolated one-atom fragment); ligand records were therefore rewritten with a placeholder three-character residue name before conversion. Proteins and ligands were converted to PDBQT with Open Babel (12), and docking search boxes were centered on each active-site ligand’s heavy-atom centroid (22 Å cubic box).

To assess whether the deposited ligand pose itself is well supported by the experimental map – an alternative explanation for a redocking failure that is independent of the docking protocol – we retrieved the official wwPDB validation report (XML) for 9L40 and 9L4B from the same AWS mirror. For the active-site ligand copy used as the redocking reference in each structure, the real-space correlation-based Q-score (13) and fractional residue inclusion (the proportion of the ligand’s atoms with adequate density support) were: 9L40/A2701, Q-score 0.553, residue inclusion 0.818; 9L4B/A2701, Q-score 0.528, residue inclusion 0.967. The Q-scores are comparable (if anything marginally higher for 9L40), so the deposited berzosertib pose is not obviously a poorly resolved outlier by that metric; residue inclusion is nonetheless visibly lower for 9L40, indicating a somewhat less complete density match for that pose. This does not, by itself, explain a 2+ Å redocking discrepancy, but it is reported here in full rather than omitted, since it bears on how confidently the redocking result should be interpreted (see Discussion). For reference, the two allosteric-site copies in 9L40 (A2702, B2702) score markedly lower still (Q-score 0.385/0.422, residue inclusion 0.364/0.485), consistent with the dimer-interface pocket being a more weakly resolved, peripheral region of the map.

Seed compound library

Five compounds formed the seed library (Table 1): the three active-site ligands extracted directly from the co-crystal structures above (berzosertib, camonsertib, and the 2YM8 CHK1 inhibitor), plus prexasertib and ceralasertib, whose canonical SMILES were obtained via literature search and independently cross-checked against reported molecular formula/exact mass (both matched exactly: prexasertib C₁₈H₁₉N₇O₂, exact mass 365.16; ceralasertib C₂₀H₂₄N₆O₂S). The PubChem and ChEMBL REST APIs used by this pipeline’s programmatic seed-retrieval script were not reachable from the compute environment used for this study, so this smaller, directly-verified seed set was used in their place; all five structures were confirmed valid and standardized with RDKit (14) (salt stripping, charge neutralization, canonicalization).

Derivative generation and developability filtering

The seed set was fragmented with the BRICS algorithm (10) into 22 unique fragments, recombined (BRICS.BRICSBUILD, RDKit), and capped at 5,000 generated candidates (a combinatorial cap, not a natural stopping point – see Limitations). Candidates were pre-filtered by a relaxed Lipinski rule (≤ 1 violation of molecular weight ≤ 500 , cLogP ≤ 5 , H-bond donors ≤ 5 , H-bond acceptors ≤ 10) (15), Ve-

Table 1: Seed compound library. Compounds marked * were extracted directly from the co-crystallized active-site ligand of the named PDB structure; the remaining SMILES were obtained via literature search and independently verified against reported molecular formula and exact mass.

Compound	Target	Provenance
Berzosertib (VE-822)	ATR	PDB 9L40, active-site ligand*
Camonsertib (RP-3500)	ATR	PDB 9L4B, active-site ligand*
CHK1 inhibitor (YM8)	CHK1	PDB 2YM8, active-site ligand*
Prexasertib	CHK1	Literature SMILES, formula-verified
Ceralasertib (AZD6738)	ATR	Literature SMILES, formula-verified

ber’s rule (rotatable bonds ≤ 10 , TPSA ≤ 140 Å²) (16), and removal of any structure matching the RDKit PAINS (17) or Brenk structural-alert catalogs, leaving 786 candidates. Every retained candidate (plus the 5 seeds) was then scored against four descriptor-based drug-likeness rules (Lipinski, Veber, Ghose (18), Egan (19)); 680 of 791 compounds satisfied $\geq 3/4$ rules with no structural alert (“developability-passed”). All values from this stage are rule-based predictions, not measurements.

Redocking validation

Before any candidate was scored, each receptor was validated by extracting its own active-site ligand, redocking it into the identical receptor/box with AutoDock Vina (9) (exhaustiveness 32, three independent seeded replicates: 1000, 1001, 1002), and computing the root-mean-square deviation (RMSD, heavy atoms) between the best-scoring pose and the native crystallographic pose, against a pre-specified pass threshold of ≤ 2.0 Å (8), fixed before any receptor was tested. Two RMSD variants are reported: a naive, index-matched RMSD (valid here because the docked pose and the input ligand file share atom order by construction within a single Vina run), and a symmetry-corrected RMSD computed with spyrmsd (20) via graph-isomorphism atom matching, since naive index-matched RMSD can overstate pose error for ligands with internal topological symmetry (e.g. equivalent ring or sulfone atoms). Before applying either metric to real poses, both were checked against two controls: a molecule scored against its own unperturbed coordinates (must return exactly 0.00 Å), and a molecule subjected to a known, exact 1.0 Å rigid translation (must return exactly 1.00 Å); both controls passed for all three ligands (Results). A receptor failing the redocking control in any replicate is excluded from candidate scoring, with the reason stated, rather than retried until it passes. Search reproducibility (pairwise RMSD between replicate best poses, independent of whether that pose is correct) was reported separately from redocking accuracy (RMSD to the native pose), since these answer different questions.

Pilot library docking

The 5 seed compounds plus the top 35 developability-passed derivatives (ranked by number of rules satisfied, then by molecular weight ascending) were docked against every receptor structure that passed redocking validation, using AutoDock Vina at exhaustiveness 16 (three independent seeded replicates: 2000, 2001, 2002) – a lower exhaustiveness than the 32 used for redocking validation itself, for compute-cost reasons. Because the central claim of this pilot is that a receptor must be validated under the protocol it will actually be screened with, this discrepancy is a real weakness in the original design, and we closed it directly rather than assert the two protocols were equivalent: the redocking validation was re-run in full (three seeded replicates, same seeds 1000/1001/1002) at exhaustiveness 16, and the same pass/fail pattern held (Results) – 9L40 still fails (3.01 ± 0.06 Å), 9L4B and 2YM8 still pass (0.59 ± 0.02 Å and 0.92 ± 0.01 Å) – so the reliability conclusion does not depend on which of the two exhaustiveness settings is used. Ligands were 3D-embedded and protonated at pH 7.4 with Open Babel. The best (most

negative) affinity across replicates is reported per compound alongside the mean and standard deviation; the mean is the primary statistic discussed in Results, since the best-of-three is a biased (downward) estimator.

Computational environment

All calculations were run with AutoDock Vina 1.2.7 (Python bindings) and Open Babel (both via pip, CPU only, no GPU available in this environment – see Limitations regarding molecular dynamics), RD-Kit 2026.03.6, and Biopython 1.88. Receptor structures were retrieved from the RCSB PDB AWS Open Data mirror snapshot dated 2026-01-01. All code and intermediate data are available at the repository accompanying this submission (see Availability of data and materials).

Results

Redocking validation: one of two ATR structures is unreliable for docking

Both RMSD sanity controls passed for all three ligands before any real pose was evaluated: crystal-vs-itself returned exactly 0.000 Å (naive and symmetry-corrected alike), and a deliberately applied, exact 1.0 Å rigid translation was recovered as exactly 1.000 Å in every case.

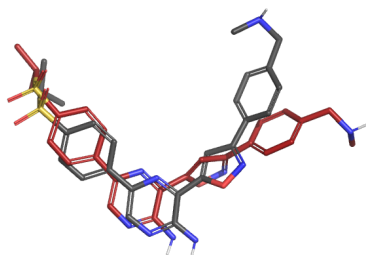
Table 2 summarizes the redocking validation across all three receptor structures over three independent seeded replicates, reporting both the naive index-matched RMSD and the symmetry-corrected RMSD (syrmsd (20)). 9L4B (ATR, camonsertib-bound, 3.20 Å) and 2YM8 (CHK1) both passed with a substantial margin under either metric (symmetry-corrected mean RMSD 0.61 ± 0.02 Å and 0.76 ± 0.01 Å respectively, well under the 2.0 Å threshold in every replicate). 9L40 (ATR, berzosertib-bound, 2.87 Å – the *higher*-resolution of the two ATR structures) failed reproducibly under both metrics: naive RMSD 2.96 ± 0.06 Å, symmetry-corrected 2.53 ± 0.01 Å. Symmetry correction narrows the gap – as expected, since berzosertib has more internal topological symmetry than camonsertib or the CHK1 ligand – but does not close it: 2.53 Å still exceeds the 2.0 Å threshold by a comfortable margin, so the failure is not a symmetry artifact.

This was also not simply search noise, but the relevant comparison is relative, not absolute: the pairwise RMSD between 9L40’s own replicate best poses (search reproducibility, 0.56 Å mean) is 12× looser than 9L4B’s (0.045 Å) and 7× looser than 2YM8’s (0.078 Å). Read against the other two structures rather than against the 2.0 Å pass threshold, 9L40’s search is both less accurate *and* less reproducible than the structures that passed – consistent with a rougher or flatter local score landscape in that pocket – though it is still far tighter than the pose-to-pose error itself, meaning Vina is converging on a consistent answer, just not the deposited one. Two further checks rule out two specific alternative explanations for the 9L40 failure (Methods): the box could not have reached the nearby allosteric pocket (60.65 Å away, far outside a 22 Å box), and the deposited ligand pose is not an obvious modeling outlier by map-model fit (Q-score comparable to 9L4B’s, though with somewhat lower residue inclusion – Methods). Per the pre-specified rule, 9L40 was excluded from candidate scoring; the pilot library docking below uses 9L4B only. Figure 1 shows the native and redocked poses for both structures directly. Because the pilot library was screened at a lower exhaustiveness (16) than used for validation above (32) for compute-cost reasons, we additionally re-ran the full redocking validation at exhaustiveness 16 (Methods) to confirm this was not a hidden inconsistency: the same pass/fail pattern held exactly – 9L40 3.01 ± 0.06 Å (fail), 9L4B 0.59 ± 0.02 Å (pass), 2YM8 0.92 ± 0.01 Å (pass) – so the reliability conclusion is not an artifact of the exhaustiveness setting used for validation.

Table 2: Redocking validation across three independent seeded Vina replicates (exhaustiveness 32). Resolution taken directly from each structure’s `_em_3d_reconstruction.resolution` mmCIF field.

Structure	Resolution (Å)	Naive RMSD (Å)	Symm. RMSD (Å)	Passes ≤ 2.0 Å	Search repro. (Å)
9L40 (ATR, VE-822)	2.87	2.96 ± 0.06	2.53 ± 0.01	No (all 3)	0.56
9L4B (ATR, camonsertib)	3.20	0.61 ± 0.02	0.61 ± 0.02	Yes (all 3)	0.045
2YM8 (CHK1)	2.07	0.90 ± 0.01	0.76 ± 0.01	Yes (all 3)	0.078

A: 9L40 (berzosertib), FAILED, RMSD 2.53 Å



B: 9L4B (camonsertib), PASSED, RMSD 0.61 Å

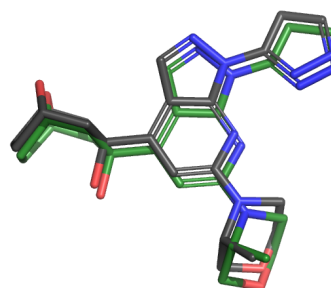


Figure 1: Native (crystal, grey) vs. redocked (colored) best pose. (A) 9L40/berzosertib: the isopropylsulfonylphenyl and aminomethylphenyl termini are visibly displaced/rotated relative to the native pose (symmetry-corrected RMSD 2.53 Å). (B) 9L4B/camonsertib: near-complete overlay (symmetry-corrected RMSD 0.61 Å).

Seed and derivative library

The 5-compound seed set yielded 22 unique BRICS fragments, from which 5,000 recombination candidates were generated (capped) and 786 passed the initial Lipinski/Veber/PAINS/Brenk pre-filter. Of the resulting 791 compounds (786 derivatives + 5 seeds), 680 (86%) satisfied $\geq 3/4$ developability rules with no PAINS/Brenk structural alert.

Pipeline sanity-check screen against the validated ATR structure

This section is a sanity check on the pipeline’s behavior, not a prospective screening result: as the analysis below shows, its outcome (the seed compound outranking its own derivatives) is close to guaranteed by construction (see Discussion), and the compound-level ranking should be read accordingly.

The 40-compound pilot set (5 seeds + 35 developability-passed derivatives) was docked against 9L4B (exhaustiveness 16, three seeded replicates each; Methods discusses the exhaustiveness used for validation vs. screening). Best-pose predicted affinity across the set ranged from -9.80 to -7.82 kcal/mol; leading with the more robust statistic, mean affinity (across 3 replicates) across all 40 compounds was -8.66 ± 0.45 kcal/mol. Table 3 shows the top 10 ranked compounds by mean affinity. Camonsertib – the receptor’s own co-crystallized ligand – ranked first, as expected for a structure the receptor was solved with. No BRICS derivative exceeded the best seed compound’s predicted affinity; the closest, deriv_002 (-9.46 ± 0.01 kcal/mol), came within 0.32 kcal/mol of camonsertib – a gap well within Vina’s reported absolute error against experiment (commonly cited at 2–3 kcal/mol), so this ranking should be read as indistinguishable rather than as a near-miss. The three top-ranked derivatives (deriv_002, deriv_003, deriv_019) share camonsertib’s oxabicyclic-ether/pyrazolopyrazine core with a single substituent varied (methylamino, methoxy, and dimethylamino respectively, in place of camonsertib’s own morpholine substituent) – an interpretable and expected outcome of BRICS recombination re-using the best-scoring seed compound’s own fragments in a new context, rather than an independent discovery of novel chemical matter. The 35-derivative subset was selected by rule count then molecular weight ascending (Methods), which systematically favors smaller ligands – themselves typically weaker Vina scorers – so the pilot set is, by its own selection rule, stacked against derivatives beating the (larger) seed compounds.

Table 3: Top 10 ranked compounds from the 40-compound pilot docking against 9L4B (mean $\pm$ SD and best across 3 seeded replicates, exhaustiveness 16).

Compound	Source	Mean affinity (kcal/mol)	Best affinity (kcal/mol)
camonsertib (seed)	seed	-9.78 ± 0.01	-9.80
deriv_002	derivative	-9.46 ± 0.01	-9.47
ceralasertib (seed)	seed	-9.39 ± 0.01	-9.40
deriv_003	derivative	-9.33 ± 0.01	-9.34
deriv_019	derivative	-9.25 ± 0.03	-9.29
deriv_032	derivative	-9.06 ± 0.17	-9.15
deriv_022	derivative	-9.08 ± 0.02	-9.10
berzosertib (seed)	seed	-8.94 ± 0.05	-8.99
deriv_033	derivative	-8.96 ± 0.02	-8.98
deriv_026	derivative	-8.94 ± 0.00	-8.94

Discussion

The central finding of this pilot is methodological rather than chemical: **a newly deposited, higher-resolution structure is not automatically a reliable docking receptor, and only an explicit redocking control establishes this, per structure, before any candidate is trusted.** 9L40 and 9L4B were solved in the same study, yet only one reproducibly passed a standard accuracy control under an otherwise identical protocol – and it is the *lower*-resolution structure (9L4B, 3.20 Å) that passed, while the higher-resolution one (9L40, 2.87 Å) failed. That inversion is, in a single sentence, the practical argument for running this control rather than trusting resolution or recency as a proxy for it. A screening campaign that skipped this control and used 9L40 by default (e.g., because it was the first hit in a search, or because it captures the more novel allosteric site) would have scored every candidate against a receptor conformation the docking protocol itself cannot reliably reproduce.

We tested, and ruled out, two specific alternative explanations rather than leaving the failure as an unresolved binary. It is not a symmetry artifact of naive RMSD: symmetry-corrected RMSD (spyrmsd) narrows the gap from 2.96 to 2.53 Å but 9L40 still fails by a comfortable margin, and the same pass/fail pattern is exactly reproduced by the two structures that were never in question (9L4B, 2YM8), which rules out a systematic RMSD computation error as well. It is not the box reaching into the emptied allosteric pocket: that pocket's nearest copy is 60.65 Å from the active site, far outside the 22 Å box used here. The remaining candidate explanations are a genuinely less well-supported deposited pose (partially, not fully, consistent with the wwPDB validation data – comparable Q-score but lower residue inclusion than 9L4B) and a genuine limitation of rigid-receptor Vina docking against this particular conformational state (e.g., an induced-fit effect specific to the dimer-interface-engaged assembly). This pilot cannot fully separate those two remaining explanations, and does not attempt to; the practical implication for anyone reusing these structures is the same either way: validate before you screen. What it does establish, with the alternative explanations we could cheaply test now closed off, is that the effect is real and not an artifact of how it was measured.

This finding rests on a single failing structure out of two tested, and a generalizable claim – that structure-dependent docking reliability is common, or that it is unrelated to resolution – would need a survey across many more recently deposited structures, not an anecdote from one. We do not make that broader claim here. What this pilot demonstrates is that the validate-before-screening control is inexpensive, automatable (the same script validated all three structures here), and caught a real, non-obvious failure on the very first pair of structures it was pointed at; turning that into a rate (what fraction of recent cryo-EM kinase depositions pass a standard redocking control, and does failure correlate with resolution, deposition year, or map-quality metrics) is the natural next study, using the same pipeline at a larger *n*.

The pilot docking result is, appropriately for a 35-derivative pilot, modest rather than dramatic: no generated derivative exceeded the predicted affinity of the seed compounds, and the best-performing derivatives are recognizable as close analogs of camonsertib rather than structurally novel chemotypes. This is consistent with a BRICS recombination run over a 5-compound seed set and a fragment pool of only 22 pieces: the search space explored is inherently local to the seed chemistry, and a pilot at this scale should not be expected to surface a structurally distinct chemotype with superior predicted affinity. What this pilot does show is that the pipeline behaves sensibly at small scale – it recovers the seed compound's own high rank (unsurprising, since camonsertib is the receptor's native ligand, but a useful sanity check that the docking protocol is not producing arbitrary rankings) and produces derivatives whose structure-score relationship is interpretable (small, defined substituent changes at one position, with correspondingly small score changes) rather than erratic. Whether a larger seed set, a larger fragment pool, or multiple recombination rounds would surface derivatives that improve on the seed compounds is an open question this pilot's scale cannot answer.

Limitations

This is a small pilot, and its scope was set by what could be executed directly and verifiably, not by what would be most impressive to report. Specifically: (1) the central reliability finding rests on **n=1** failing structure out of two tested; it demonstrates that the validate-before-screening control is necessary and can catch a real, non-obvious failure, but a claim about how *common* such failures are, or what they correlate with, would need a survey across many more structures (Discussion) and is not made here; (2) the BRICS recombination cap (5,000) is combinatorial, not principled, and a larger or differently seeded run would sample a different, likely larger, region of the accessible chemical space; (3) the 40-compound pilot library is a small, deterministic subset of the developability-passed derivatives (ranked by rule count then molecular weight – a rule that, incidentally, is stacked against derivatives outscoring the larger seed compounds), not a comprehensive screen, and enrichment-style retrieval statistics (ROC-AUC, enrichment factor) are not meaningful at this scale and are not reported; (4) no molecular dynamics or MM-GBSA relative energetic estimation was run – both require substantially more compute (GPU acceleration for MD at any informative timescale) than was available, and docking scores alone are known to correlate imperfectly with binding affinity; (5) no cross-kinase selectivity panel was run – this would require additional receptor structures, each independently redocking-validated by the same standard applied to the three structures here, which was beyond this pilot’s scope; (6) as throughout, nothing here establishes ATR/CHK1 functional modulation, inhibition, or activation – only predicted binding and ranking; (7) the seed and reference SMILES for prexasertib and ceralasertib were obtained via web search rather than a structured chemical database query (PubChem/ChEMBL were not reachable) and, while independently verified against reported molecular formula and exact mass, should be re-confirmed against a primary chemical database before further use.

Conclusions

Redocking validation – not resolution, recency, or biological interest – is what determines whether a deposited structure is a trustworthy docking receptor. Applied to two contemporaneous ATR structures from the same study, this control passed the lower-resolution one and failed the higher-resolution one, a distinction that would not have been visible without running it, that survives symmetry-corrected RMSD, and that is not explained by the box reaching an emptied allosteric site. Screened against the validated structure, a small BRICS-derived sanity-check library reproduced the expected top rank of the receptor’s own native ligand and generated interpretable, if modest, analogs of it, without yet surfacing a chemotype that improves on the seed compounds – a realistic, honestly reported outcome for a pilot of this scale, and one further reason to treat larger claims from similarly small screens with caution. The natural next step is the same pipeline at a larger *n*: a survey of redocking pass rates across recently deposited structures would turn this single, real finding into a rate. All structures, code, and intermediate data used in this study are openly available for independent verification and reuse.

Declaration of AI assistance

Portions of this study’s code, analysis, and manuscript drafting were produced with the assistance of an AI coding assistant (Claude, Anthropic), operating under direct human review and direction throughout, including the verification steps described in Methods (e.g., cross-checking externally sourced facts against primary sources such as mmCIF structure files, wwPDB validation reports, and independently reported molecular formulae before use). All computational results reported are from actual executed code, not AI-generated approximations; all data-processing and analysis code is available at the repository in Availability of data and materials for independent verification.

Availability of data and materials

All code, the seed and derivative compound tables, redocking validation results, and pilot docking results are available at <https://github.com/ethanyyy123/research-molecule-editing> (branch [claude/cadd-fragile-sites-optimization-0u9fsp](#) at the time of this submission; an archived, versioned release should be minted, e.g. via Zenodo, before final submission). Receptor structures are publicly available from RCSB PDB (9L40, 9L4B, 2YM8) and were retrieved via the AWS Open Data mirror ([s3://pdbsnapshots](#)).

Competing interests

[PLACEHOLDER – e.g. “The authors declare no competing interests,” if accurate; confirm before submission.]

Funding

[PLACEHOLDER – funding source(s), or “Not applicable” if self-funded/no external support.]

Authors’ contributions

[PLACEHOLDER – e.g. following CRediT roles: conceptualization, methodology, software, formal analysis, investigation, writing.]

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[PLACEHOLDER – optional; e.g. acknowledgement of compute resources or advisors, if applicable.]

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